

Evaluation and Treatment of Hypertensive Urgencies and Emergencies

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Severe elevations in blood pressure (BP) can present as hypertensive emergencies or hypertensive urgencies.^{1–3} A *hypertensive emergency* is defined as severe elevation in BP associated with evidence of new or worsening target-organ damage, such as coronary ischemia, dissecting aortic aneurysm, pulmonary edema, acute kidney failure, papilledema, hypertensive encephalopathy, cerebral hemorrhage, and eclampsia. In these cases, both the rate and magnitude of an increase in BP are at least as equally important to the absolute BP level in determining the magnitude of organ injury. As such, although in most of these cases BP is usually greater than 180/120 mm Hg,^{1,2} there is no specific BP threshold to define hypertensive emergencies, and previously normotensive subjects can present with hypertension-mediated organ damage at lower BP levels.³ Hypertensive emergencies require immediate and careful intervention to reduce BP (not necessarily to normal), preferably with intravenous (IV) drug therapy in an intensive care unit (ICU) to limit further target-organ damage.^{1–3} A *hypertensive urgency* is the clinical situation of a patient who presents with severe BP elevation without acute or impending target organ dysfunction. These patients require a gradual BP reduction within hours with oral medications and do not normally require hospitalization.^{1,2,4}

The term *malignant hypertension* first appeared in 1928 and described patients with extremely high BP values to emphasize that because of rapid target organ damage, their average prognosis was similar to that of most cancer patients.⁵ Subsequently, malignant hypertension is used to describe patients with greatly elevated BP and vascular damage that manifests as the bilateral presence of retinal flame hemorrhages and/or papilledema, which can be accompanied by encephalopathy, acute kidney injury, or thrombotic microangiopathy.^{2,3,6} Dramatic advances in both in-hospital and outpatient treatment have led to an improved prognosis of such cases and a decrease in 1-year mortality from 80% in 1928 to 50% in 1955 and to less than 10% after 1990.^{5,7}

ETIOLOGY AND PATHOGENESIS

Hypertensive emergencies and urgencies can develop *de novo* in normotensive individuals but usually complicate underlying primary or secondary hypertension.^{4,8} Box 38.1 shows the most common clinical situations identified as hypertensive emergencies, along with cases of severe BP elevation without target-organ damage. In some hypertensive emergencies, an underlying condition is the clear cause of acute BP elevation. For example, in acute glomerulonephritis, renal crisis in patients with systemic sclerosis, and renal artery stenosis, severe BP elevations usually result from increased activity of the renin-angiotensin system (RAS). In pheochromocytoma, cocaine intoxication, or spinal cord injury, acutely elevated BP is the result of excess catecholamine release. In other patients, acute sustained elevations in BP itself are the etiologic factor, resulting in conditions such as hypertensive

encephalopathy or severe hypertension with acute left ventricular failure and pulmonary edema. In some cases, however, it may be difficult to differentiate whether BP elevation is the cause or the result of a hypertensive emergency. For example, in a patient with intracerebral hemorrhage, an acute marked BP increase may be the primary cause; alternatively, a hemorrhage of other etiology (i.e., coagulation deficit) may have occurred, followed by BP elevation to preserve cerebral tissue blood supply. Thus, a careful diagnostic evaluation of hypertensive emergencies and urgencies is essential to guide proper treatment.

The most common clinical setting for hypertensive urgency is a patient with chronic hypertension (often undiagnosed, untreated, or uncontrolled) whose usual BP is greater than 180/120 mm Hg. In many of these patients, chronically elevated BP does not affect target organ perfusion because of *autoregulation*, which is the ability of blood vessels to dilate or constrict in response to changes in perfusion pressure and thereby maintain normal organ perfusion.^{9,10} In the brain, autoregulation primarily preserves blood flow, whereas in the kidney, it preserves glomerular hydrostatic pressure. Arteries from normotensive individuals can maintain flow over a wide range of mean arterial pressures (70–170 mm Hg, reflecting systolic blood pressure [SBP] of around 90–200 mm Hg). Chronic BP elevations cause compensatory functional and structural changes in the arteriolar circulation and shift the autoregulatory curve to the right, which allows hypertensive patients to maintain normal perfusion and avoid excessive blood flow at higher BP levels (Fig. 37.1).^{9,10} As such, most patients with chronically elevated BP greater than 180/120 mm Hg visiting the emergency department (ED) have no evidence of acute target-organ damage, and thus have hypertensive urgencies rather than emergencies.

The factors that lead to the severe and rapid BP elevation causing a hypertensive emergency are poorly understood. In some cases, the rapidity of the onset suggests a triggering factor (i.e., release of a humoral vasoactive factor) superimposed or not on preexisting hypertension. Most often, the hypertensive emergency is likely again a nonspecific consequence of chronically elevated BP, as with time, further structural and functional changes in cerebral and renal arterioles can lead to a progressive inability of the arterioles to autoregulate properly.^{9–12} Calcium channel blockers (CCBs) may also paradoxically increase glomerular pressure because of their ability to impair the renal autoregulatory response if systemic BP is not controlled. One way or another, target organ damage occurs when the autoregulatory mechanisms cannot maintain normal perfusion pressures in certain vascular beds (especially the retina, brain, and kidney) when systemic BP rises markedly.^{11–13} This results in high-shear forces causing extensive endothelial injury, with detachment and exposure of subendothelium to blood, and with activation of platelets and the coagulation cascade, leading to fibrin formation, increased vascular wall permeability, and vascular smooth muscle cell proliferation culminating in vasculopathy, occasionally with fibrinoid necrosis. Thrombotic microangiopathy may result after formation of platelet-rich

BOX 38.1 Common Hypertensive Emergencies and Urgencies

Acceleration of Chronic Hypertension

- With papilledema ± encephalopathy ± kidney failure
- With features of thrombotic microangiopathy (thrombocytopenia, hemolytic anemia, renal failure, ± papilledema, ± encephalopathy)

Cardiovascular Conditions

- Acute myocardial ischemia/infarction caused by coronary artery disease
- Acute left ventricular failure/pulmonary edema
- Acute aortic dissection
- Severe hypertension after coronary bypass or other vascular surgery

Kidney Conditions

- Acute or rapidly progressive glomerulonephritis
- Renovascular hypertension
- Renal crises from scleroderma or collagen vascular disease
- Severe hypertension after kidney transplantation

Neurologic Conditions

- Hypertensive encephalopathy
- Intracerebral hemorrhage
- Subarachnoid hemorrhage
- Cerebral embolism or atherothrombotic cerebral infarction
- Severe hypertension after thrombolysis for atherothrombotic stroke
- Acute head trauma
- Guillain-Barré syndrome

Excess Circulating Catecholamine Conditions

- Pheochromocytoma crisis
- Interactions of tyramine-containing foods with monoamine oxidase inhibitors
- Rebound hypertension after sudden withdrawal of centrally acting α_2 -agonists (clonidine, methyldopa, or other)
- Use of sympathomimetic drugs (phencyclidine, phenylpropanolamine, cocaine, or other)
- Automatic hyperreflexia after spinal cord injury

Pregnancy-Related Condition

- Severe preeclampsia/HELLP syndrome
- Eclampsia

Surgical Conditions

- Severe hypertension in patients requiring immediate surgery
- Perioperative hypertension
- Epistaxis unresponsive to anterior/posterior packing

Hypertension Associated With Severe Burns

HELLP, Hemolysis, elevated liver enzymes, and low platelets.

thrombi with obliteration of the microcirculation, platelet consumption, and intravascular hemolysis because of trapping and destruction of erythrocytes within the fibrin network. Activation of hormonal systems and release of vasoactive substances (RAS, catecholamines, endothelin, vasopressin) may create a vicious cycle in which elevated BP can both cause and result from vascular injury.^{3,4,13,14} In patients with hypertensive emergency, the typical structural kidney changes are onion-skin appearances of small arteries and arteriolar and glomerular collapse (Fig. 38.2) with electron-lucent widening of the subendothelial zone and wrinkling of the glomerular capillary walls, whereas fibrinoid necrosis of small arteries is less common.¹⁵

Renal Autoregulation

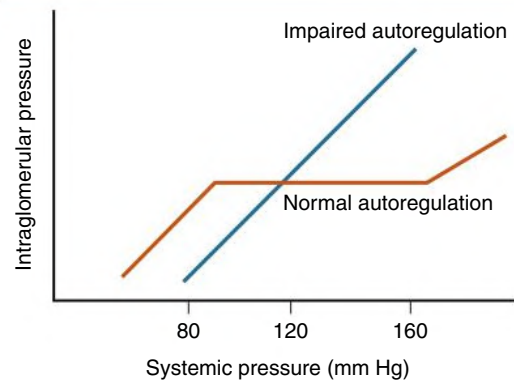


Fig. 38.1 Renal Autoregulation. Relationship of systemic to glomerular pressure in the setting of normal or abnormal renal autoregulation.

In normotensive or minimally hypertensive individuals, such as children or pregnant women, the symptoms and signs of a hypertensive emergency may occur at lower BP levels than in hypertensive patients because adaptive chronic microvascular changes are absent. In such cases, the rate and magnitude of BP increase are more important than the actual BP level, which may be lower than the commonly used threshold of 180/120 mm Hg.¹²

EPIDEMIOLOGY

The incidence of hypertensive emergencies and urgencies can be estimated from large observational studies of patients with hypertension. Among 333,407 patients visiting 10 EDs in Italy over a 1-year period, 1155 had a hypertensive urgency, defined as BP at or greater than 220/120 mm Hg (3.46/1000/year), and 391 patients had a hypertensive emergency (1.17/1000 visits/year).¹⁶ The most common manifestations of hypertensive emergency were acute pulmonary edema (30.9%), stroke (22%), myocardial infarction (MI; 17.9%), acute aortic dissection (7.9%), acute kidney failure (5.9%), and hypertensive encephalopathy (4.9%). A systematic review of seven studies with variable definitions reported an incidence of hypertensive urgency between 1.64 and 24 patients/1000 ED visits/year and of hypertensive emergency between 1.17 and 7.6 patients/1000 ED visits/year.¹⁷ Furthermore, a wide-scale retrospective analysis of the Nationwide Emergency Department Sample (NEDS), representing about 20% of all hospital-based ED visits in the United States, showed that both hypertensive emergencies and urgencies may be increasing. Between 2006 and 2013 the incidence of acute hypertension (diagnosed by International Classification of Diseases [ICD]-9 Clinical Modification [CM] coding) increased from 1.8 to 4.6/1000 ED visits/year, and the incidence of hypertensive emergencies increased from 0.7 to 1.7/1000 visits/year.¹⁸ There is also evidence that hypertensive urgencies and emergencies are more common in middle- and low-income countries,^{19,20} and in deprived urban populations and ethnic minorities of high-income countries,²¹ that is, in patients who often have poor BP control.

People with a history of chronic kidney disease, coronary artery disease (CAD), or stroke also are at higher risk of hypertensive crises. Hypertensive emergency is more common in men, older individuals, and those with diabetes and hyperlipidemia.²² Importantly, 24% of patients presenting with a hypertensive crisis are suggested to be completely nonadherent (none of the prescribed drugs detected

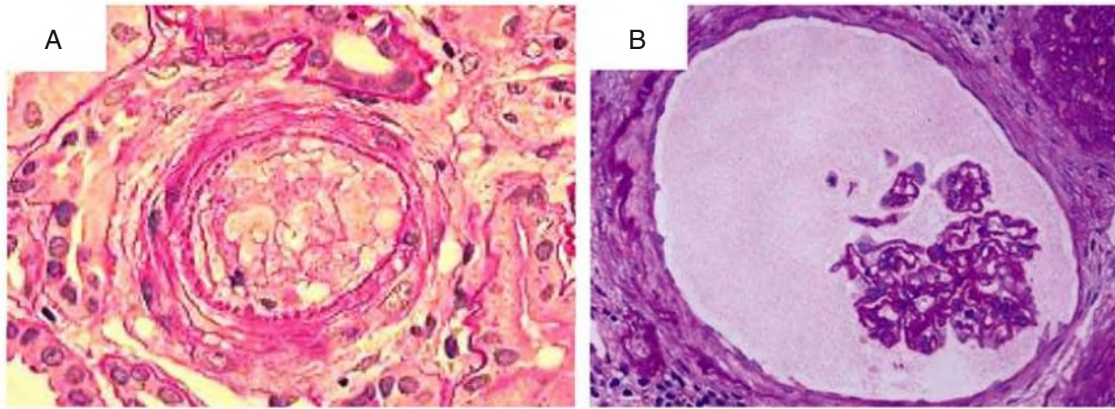


Fig. 38.2 Common Kidney Biopsy Findings in Patients With Hypertensive Emergency-Related Renal Dysfunction. (A) Concentric subendothelial edematous thickening (“onion skin” appearance) of an arteriole. (B) Collapsed glomerulus (PAS stain). (From Nonaka K, Ubara Y, Sumida K, et al. Clinical and pathological evaluation of hypertensive emergency-related nephropathy. *Intern Med.* 2013;52:45–53.)

in urine) and another 34% partially nonadherent to their prescribed treatment.²³ Another study in patients presenting with hypertensive urgency showed that 38% had skipped antihypertensive medications and 45% had pain of noncardiac origin.²⁴

Hypertensive emergencies are associated with an increased risk for mortality, with an acute in-hospital mortality of 11% to 12.5% and a 1-year mortality of 38.9%. Mortality was greater in those who experienced neurologic or cardiovascular emergencies.^{25,26} As expected, mortality associated with hypertensive urgency is lower (in-hospital mortality between 0.3%– and .8% and 1-year mortality of 8.9%).^{25,26} With regards to kidney function, about 70% of patients with malignant hypertension have an estimated glomerular filtration rate (eGFR) of less than 60 mL/min/1.73m² at presentation; among these patients, 7% would need permanent kidney replacement therapy and another 20% to 25% would develop end-stage kidney disease (ESKD), requiring dialysis over long-term follow-up.^{27,28}

DIAGNOSTIC EVALUATION

The primary goal of the diagnostic process is to differentiate a true hypertensive emergency from a hypertensive urgency because of the different therapeutic approaches. The second goal is rapid assessment of the type and severity of ongoing target-organ damage. In some hypertensive emergencies, the history (e.g., acute head trauma, pre-eclampsia, scleroderma) or overt symptoms and signs (e.g., chest/back pain, dyspnea, focal neurologic deficit, throbbing abdominal mass) may quickly guide the diagnosis; in patients with more general symptoms (headache, visual disturbances, dizziness, nausea, or altered mental status), however, the evaluation must be more comprehensive.

The diagnostic approach begins with the patient's history, with attention to duration, severity, and treatment of preexisting hypertension and associated conditions, and to adherence to prescribed agents and possible use of substances that increase BP^{2–4,8} (Box 38.2). BP readings should be performed in both arms and a leg to detect BP differences caused by aortic dissection and, if possible, in both sitting and standing positions.³ Repeated BP measurements should be performed at regular intervals because in one-third of patients with hypertensive urgencies, BP will fall considerably without antihypertensive drugs.²⁹ Physical examination of cardiac, pulmonary, peripheral vascular, and neurologic systems with assessment of mental status should follow. Funduscopic (ophthalmoscopic) examination should look for hemorrhages, exudates (cotton wool spots), and papilledema as a crucial part of the diagnostic process.^{4,8,12}

BOX 38.2 Diagnostic Evaluation for Hypertensive Emergencies and Urgencies

History

- Previous diagnosis and treatment of hypertension
- Symptoms, previous diagnoses, and treatment of cardiac, cerebral, renal, and visual damage
- Intake of pressor agents: sympathomimetics, illicit substances

Blood pressure

Physical examination

- Cardiac
- Vascular
- Pulmonary
- Neurologic
- Optic fundi

Laboratory studies

- Complete blood count, creatinine, urea, electrolytes, urinalysis,
- Specific tests by indication: fibrinogen, LDH, schistocytes, urine microscopy for red blood cells and casts, urine albumin/creatinine ratio, troponin, CK-MB, NT-proBNP, pregnancy test
- Plasma renin activity, aldosterone, and catecholamines if secondary hypertension is suspected

12-Lead electrocardiogram

Chest radiograph

Further investigations (according to clinical presentation)

- Kidney ultrasound including assessment of renal artery flow with Doppler
- Brain CT scan or MRI
- Transthoracic echocardiography
- Thoracoabdominal CT scan or MRI

CK-MB, Creatinine kinase-MB fraction; CT, computed tomography; LDH, lactate dehydrogenase; MRI, magnetic resonance imaging; NT-proBNP, N-terminal pro-B natriuretic peptide.

This initial examination should not miss signs of secondary hypertension. An abdominal bruit may indicate renovascular hypertension; a pulsatile abdominal mass suggests abdominal aneurysm; palpable kidneys may be because of polycystic kidney disease; a radial-femoral pulse delay suggests aortic coarctation; abdominal striae and central obesity are observed with Cushing syndrome; and exophthalmos may indicate hyperthyroidism.

Initial laboratory studies in a hypertensive emergency include a complete blood count (CBC) with peripheral smear, urinalysis,

creatinine and urea concentrations, and electrolytes.^{4,8,12} Comparison of kidney function with a patient's recent measurement is important. Further tests should be guided by clinical suspicion.² High troponin and creatine kinase-MB (CK-MB) or high N-terminal pro-hormone B-type natriuretic peptide (NT-proBNP) can guide the diagnosis of acute MI or decompensated heart failure (HF). Acute deterioration in kidney function, microscopic hematuria with red blood cell casts, or nephritic urine sediment suggest acute glomerulonephritis. Features of hemolytic anemia and thrombocytopenia characterize patients with malignant hypertension-associated TMA. Differentiating hypertension-induced TMA from thrombotic thrombocytopenic purpura (TTP) and hemolytic-uremic syndrome (HUS) can be difficult.^{3,30} The coexistence of severe BP elevation and advanced retinopathy are helpful, along with the fact that malignant hypertension is usually accompanied by less pronounced thrombocytopenia and fewer schistocytes present in a peripheral blood smear than TTP and HUS; however, some cases would require measurement of ADAMTS13 and other tests.

If a secondary form of hypertension is suspected, samples for plasma renin activity, aldosterone concentration, and plasma-free catecholamines and metanephrines should also be drawn *before* initiation of treatment with the patient supine. If the patient is already receiving RAS inhibitors, diuretics, or β -blockers, the diagnostic accuracy of these tests changes considerably, and results should be interpreted with caution.

Electrocardiography to rule out myocardial ischemia and left ventricular strain or hypertrophy, as well as chest radiography, should be performed in every patient. Kidney ultrasound, with renal artery Doppler, if available, is also useful to rule out abnormalities such as differences in kidney size or perfusion,³¹ especially in patients with reduced eGFR or with abnormalities on urinalysis.

Neurologic syndromes associated with hypertension, including subarachnoid hemorrhage, intracerebral hemorrhage, thrombotic stroke, and hypertensive encephalopathy, are difficult to distinguish from one another only by means of clinical examination. Computed tomography (CT) or magnetic resonance imaging (MRI) provides a definite diagnosis of a hemorrhagic or thrombotic stroke. Echocardiography, thoracoabdominal CT or MRI, or abdominal ultrasound may be needed in patients with suspected aortic dissection or pheochromocytoma.²⁻⁴

TREATMENT

General Principles and BP Treatment Thresholds for Hypertensive Emergencies

No randomized controlled trials (RCTs) examine the effect on hard outcomes of treatment strategies with different speed and degree of BP reduction for hypertensive emergencies, with the exception of acute stroke. Recommended strategies are based on consensus from clinical experience and observations on intermediate outcomes.¹⁻³ Nevertheless, clinical data document that lowering BP in hypertensive emergencies is beneficial: papilledema and exudates regress, hypertensive encephalopathy vanishes, pulmonary edema resolves, and kidney function improves. On the other hand, abrupt or excessive BP lowering may cause or contribute to cerebral, coronary, or renal ischemia. Oral loading of antihypertensive agents can have cumulative undesirable effects; the most typical example is use of sublingual nifedipine with potent but unpredictable decrease in BP that may shunt blood away from the cerebral penumbra, resulting in ischemic events.³² Overall, understanding autoregulation and consideration of comorbidities, such as age and extent of vascular disease, are crucial for therapeutic decisions. Thus, the goal of therapy

is not to normalize BP rapidly but rather to limit target-organ damage by gradually reducing BP, while minimizing the risk of tissue hypoperfusion.¹¹

There are three major considerations before initiating treatment: (1) establishing whether there is a distinct cause of BP elevation (e.g., eclampsia, acute head trauma) or a distinct target-organ damage (acute MI, aortic dissection) that would require specific interventions in addition to BP lowering; (2) the timescale and magnitude of BP lowering required; and (3) the optimal type of drug treatment.² In hypertensive emergencies, IV treatment with a short-acting and easily titratable agent is recommended to allow tight control of BP response (Tables 38.1 and 38.2). This treatment may be initiated in the ED but should be continued in an ICU or a high dependency unit (HDU) to enable continuous BP/hemodynamic monitoring and continuous clinical surveillance.¹⁻³ A further consideration before initiation of IV therapy is assessment of the patient's volume status. With the obvious exceptions of patients presenting with volume overload and/or pulmonary edema, several patients with a hypertensive emergency may be volume depleted because of pressure natriuresis, and diuretics should not be used; rather, fluid administration may help restore organ perfusion and prevent a precipitous fall in BP.⁴ Diuretics should be especially avoided in hypertensive emergencies caused by catecholamine excess states (pheochromocytoma, monoamine oxidase inhibitor crisis, cocaine intoxication) because these patients are usually volume depleted.

Current guidelines have minor differences with regards to the recommended timescale and magnitude of BP fall for major hypertensive emergencies.¹⁻³ The 2017 American College of Cardiology/American Heart Association (AHA/ACC) Guidelines recommend reducing BP by a maximum of 25% within the first hour to SBP/diastolic blood pressure [DBP] levels of 160/100 to 160/110 mm Hg over the next 2 to 6 hours and to normal over the next 24 to 48 hours in most patients. In patients with severe preeclampsia or eclampsia and those with pheochromocytoma crisis, SBP should be reduced to less than 140 mm Hg within an hour and in those with aortic dissection to less than 120 mm Hg in the same timeframe.¹

The 2018 European Society of Cardiology/European Society of Hypertension (ESC/ESH) guidelines differ in recommending immediate BP reduction to less than 160/105 mm Hg in patients with severe preeclampsia or eclampsia and to SBP less than 140 mm Hg in those with acute coronary event and acute cardiogenic pulmonary edema.²

With regard to acute ischemic stroke, evidence from RCTs does not conclusively suggest a benefit from BP-lowering in short-term and midterm mortality risk and functional outcomes.³³ Therefore, current recommendations suggest that patients who are candidates for treatment with IV tissue plasminogen activator or mechanical thrombectomy should have their BP reduced to less than 185/110 before treatment.^{1-3,34} For patients that are not candidates for such interventions and have BP greater than 220/120 mm Hg, most guidelines agree that the evidence is uncertain, but it is reasonable to reduce BP by 15% within the first 24 hours.^{1,2,34} With regards to acute intracerebral hemorrhage (ICH), a previous RCT showed that lowering SBP to less than 140 mm Hg within an hour is safe and may improve functional outcome³⁵; however, a subsequent RCT that examined BP lowering within 4.5 hours of an acute ICH found that treatment to SBP levels of 110 to 139 mm Hg did not lead to better outcomes compared with treatment to 140 to 179 mm Hg and was associated with more renal adverse effects.³⁶ Thus, in patients with acute ICH presenting with SBP levels greater than 220 mm Hg, BP should be lowered using IV drugs to goal levels of 140 to 180 mm Hg.^{1,2}

After the BP has been controlled for a suitable period, typically 12 to 24 hours, which allows autoregulation to reestablish, the IV medication can gradually be reduced and replaced by oral agents. Typically, a dihydropyridine CCB, α - and β -blocker, or RAS blocker is used, depending on the suspected cause and possible ongoing investigations for secondary hypertension.¹¹

Specific Aspects of Antihypertensive Drug Use for Hypertensive Emergencies

The need for gradual and tightly controlled BP reduction requires the use of IV drugs (see Table 38.1), the effects of which can be promptly reversed if the response is excessive. Previous systematic

TABLE 38.1 Pharmacologic Agents for Treatment of Hypertensive Emergencies

Drug	Mechanism of Action	Dose	Onset of Action	Duration of Action	Adverse Effects ^a	Special Indications
Vasodilators						
Nicardipine hydrochloride	Calcium channel blocker	Initial 5 mg/h IV; increase every 5 min up to 15 mg/h	5–15 min	15–30 min, may exceed 4 h	Tachycardia, headache, flushing, nausea, vomiting, local phlebitis	Most hypertensive emergencies except acute HF
Clevidipine butyrate	Calcium channel blocker	1–2 mg/h IV; increase every 5–10 min up to 16 mg/h	2–4 min	5–15 min	Tachycardia, headache, flushing, HF deterioration	Most hypertensive emergencies; caution with severe aortic stenosis
Isradipine	Calcium channel blocker	0.15 μ g/kg/min IV, increase by 0.0025 μ g/kg/min every 15 min Maintenance infusion 0.15 μ g/kg/min	1–10 min	1–2 h	Headache, flushing, peripheral edema, dizziness, tachycardia	Perioperative, pregnancy
Fenoldopam mesylate	Dopamine-1 receptor agonist	Initial 0.1–0.3 μ g/kg/min IV; increase by 0.05–0.1 μ g/kg/min every 15 min up to 1.6 μ g/kg/min	>5 min	30 min	Tachycardia, headache, nausea, flushing	Most hypertensive emergencies; caution with glaucoma
Sodium nitroprusside	$\uparrow$ Cyclic GMP, blocks intracellular Ca^{2+} increase	Initial 0.25–0.5 μ g/kg/min IV ^b ; increase by 0.5 μ g/kg/min up to 10 μ g/kg/min	Immediate	1–2 min	Nausea, vomiting, muscle twitching, thiocyanate and cyanide intoxication, impaired cerebral autoregulation, coronary steal syndrome	Caution in situations associated with CNS manifestations, hepatic failure or renal failure; probably should be avoided if given other agents, especially fenoldopam
Nitroglycerin	$\uparrow$ Nitrate receptors	Initial 5 μ g/min IV; increase by 5 μ g/min every 3–5 min up to 20 μ g/min	2–5 min	5–10 min	Headache, vomiting, methemoglobinemia, tachyphylaxis, tolerance with prolonged use	Coronary ischemia, pulmonary edema
Enalaprilat	ACE inhibitor	Initial 1.25 mg over 5 min IV; increased up to 5 mg every 6 h	15–30 min	6–12 h	Precipitous fall in BP in high-renin states, variable response, acute renal failure	Acute left ventricular failure; avoid in acute MI
Hydralazine hydrochloride	Opens K^+ channels	Initial 10–20 mg IV; repeat every 4–6 h	10–20 min	1–4 h	Tachycardia, flushing, headache, vomiting, aggravation of angina	Must be given with concomitant IV β -blockers to avoid precipitation of angina but <i>not</i> a preferred initial choice or treatment

Continued

TABLE 38.1 Pharmacologic Agents for Treatment of Hypertensive Emergencies—cont'd

Drug	Mechanism of Action	Dose	Onset of Action	Duration of Action	Adverse Effects ^a	Special Indications
Adrenergic Inhibitors						
Labetalol hydrochloride	α_1 -, β -Blocker	Initial 0.3–1.0 mg/kg dose (maximum 80 mg) in slow IV injection and repeat after 10 min <i>or</i> 0.4–1.0 mg/kg/h IV infusion up to 3 mg/kg/h Maximum dose 300 mg; can be repeated every 4–6 h	5–10 min	3–6 h	Nausea, vomiting, scalp tingling, bronchoconstriction, dizziness, heart block, heart failure	Most hypertensive emergencies except acute HF
Esmolol hydrochloride	β_1 -Blocker	0.5–2 mg/min IV infusion <i>or</i> 250–500 μ g/kg/min IV bolus, then 50–100 μ g/kg/min by infusion; may repeat bolus after 5 min or increase infusion to 300 μ g/min	1–2 min	10–30 min	Nausea, asthma, first-degree heart block, HF, thrombophlebitis, COPD	Aortic dissection, perioperative, increased heart output or heart rate
Urapidil	α_1 -Blocker, serotonin (5-HT _{1A}) receptor agonist	12.5–25 mg IV bolus followed by 5–40 mg/h IV infusion	3–5 min	4–6 h	Headache, dizziness	Perioperative
Phentolamine	α -Blocker	Initial 5–15 mg IV bolus; repeat after 10 min	1–2 min	10–30 min	Tachycardia, flushing, headache	Catecholamine excess

^aHypotension may occur with all agents.

^bRequires light-resistant delivery system.

ACE, Angiotensin-converting enzyme; BP, blood pressure; CNS, central nervous system; COPD, chronic obstructive pulmonary disease; GMP, guanosine monophosphate; HF, heart failure; IV, intravenous; MI, myocardial infarction.

TABLE 38.2 Management of Specific Types of Hypertensive Emergencies

Type of Emergency	First-Choice Drug(s)	Second-Choice or Additional Drug(s)	Drugs to Avoid	Aim of BP Reduction
Cardiac				
Coronary ischemia/infarction	Nicardipine, clevidipine, labetalol, nitroglycerin	Sodium nitroprusside, esmolol if heart failure absent	Diazoxide, hydralazine	Improvement in cardiac perfusion
Heart failure, pulmonary edema	Clevidipine, nitroglycerin, fenoldopam,	Sodium nitroprusside, enalaprilat; loop diuretics	Diazoxide, hydralazine; β -blockers	Decrease in afterload
Aortic dissection	Labetalol or combination of esmolol with sodium nitroprusside or fenoldopam or nicardipine		Diazoxide, hydralazine	Decrease of aortic wall stress with systolic BP reduction <120 mm Hg in 1 hour
Renal				
Acute glomerulonephritis, collagen vascular renal disease, or renal artery stenosis	Fenoldopam	Nicardipine, labetalol, clevidipine; diuretics for volume overload	Sodium nitroprusside; ACE inhibitors and ARBs	Reduction in vascular resistance and volume overload without compromise of renal blood flow or glomerular filtration rate

TABLE 38.2 Management of Specific Types of Hypertensive Emergencies—cont'd

Type of Emergency	First-Choice Drug(s)	Second-Choice or Additional Drug(s)	Drugs to Avoid	Aim of BP Reduction
Scleroderma crisis	Enalaprilat or other ACE inhibitor	Angiotensin receptor blocker, fenoldopam	Corticosteroids, ^a diuretics	Decrease BP up to 25% over 1 hour
Neurologic				
Hypertensive encephalopathy	Nicardipine, fenoldopam, labetalol, clevidipine	Nitroprusside, esmolol, urapidil		Decrease BP up to 25% over 1 hour
Ischemic stroke	Nicardipine, labetalol, clevidipine	Nitroprusside, nimodipine, esmolol, urapidil		Reduction of BP if >220/120 mm Hg by up to 15% within first 24 hours to avoid impairing cerebral blood flow in penumbra Reduction of BP if >185/110 in candidates of thrombolysis
Intracerebral hemorrhage	Nicardipine, labetalol, clevidipine	Fenoldopam, nitroprusside, esmolol, urapidil, nimodipine for subarachnoid hemorrhage		For patients presenting with SBP >220 mm Hg, it is reasonable to reduce SBP to 140–180 mm Hg. For patients presenting with SBP 150–220 mm Hg, decreasing to <140 mm Hg is not of benefit.
Catecholamine Excess States				
Pheochromocytoma	Phentolamine or labetalol	β-Blocker in the presence of phentolamine, sodium nitroprusside	Diuretics, β-blockers alone	Control of BP paroxysms from sympathetic stimulation; reduce SBP <140 mm Hg over 1 hour
Ingestion of cocaine or other sympathomimetic	Phentolamine or labetalol	β-Blocker in the presence of phentolamine, sodium nitroprusside	Diuretics	Control of BP paroxysms from sympathetic stimulation
Perioperative/Postoperative Hypertension				
Coronary artery surgery	Nitroglycerin, nicardipine, clevidipine	Esmolol, labetalol, fenoldopam, isradipine, urapidil		Protection against target organ damage and surgical complications (keep BP <140/90 mm Hg)
Noncardiac surgery	Esmolol, labetalol, fenoldopam, nicardipine, clevidipine, urapidil, nitroglycerin			Protection against target organ damage and surgical complications
Pregnancy Related				
Eclampsia	Labetalol, urapidil	Isradipine, nicardipine, MgSO ₄ , methyldopa, hydralazine	Nitroprusside, ACE inhibitors, ARBs	Reduce BP <140–160/105 mm Hg over 1 hour and protect placental blood flow

^aCorticosteroids may worsen hypertension in scleroderma renal crisis.

ACE, Angiotensin-converting enzyme; ARBs, angiotensin receptor blockers; BP, Blood pressure; SBP, systolic blood pressure.

reviews and meta-analyses showed minor differences in the degree of BP lowering and no differences in morbidity or mortality among these agents because of the relative paucity of large RCTs with appropriate follow-up.^{37,38} Thus, treatment practices were mainly empiric. More recent evidence, however, suggests some benefits of novel agents.

Nicardipine is a dihydropyridine CCB with intermediate onset and duration of effect and strong cerebral and coronary vasodilatory activity. It is useful for most hypertensive emergencies, especially in patients with CAD. Nicardipine potentiates curare effects and interacts

with inhalant anesthetics.^{4,11,39,40} In an RCT of 226 patients with acute SBP of 180 mm Hg or higher, those receiving nicardipine reached physician-specified target range slightly more often (92% vs. 83%, $P = .039$) than those receiving labetalol.⁴¹ A systematic review of studies comparing nicardipine with labetalol suggested that the two drugs had comparable efficacy and safety, but nicardipine showed more predictable and consistent BP response.⁴² Nicardipine can be particularly useful in patients with acute coronary syndromes or hypertensive emergencies involving the brain because it reduces myocardial and cerebral ischemia.

Clevidipine butyrate is an ultrashort-acting (within 1–2 minutes) third-generation CCB that acts through inhibition of extracellular calcium influx and reduces peripheral vascular resistance without affecting venous vascular tone or cardiac filling pressure. Clevidipine is rapidly hydrolyzed by blood esterases, and thus its metabolism is not affected by renal or hepatic function.^{4,39,40} In clinical studies, clevidipine was effective and safe for the control of perioperative hypertension and hypertensive emergencies. Three RCTs including more than 1500 patients with perioperative acute hypertension that compared clevidipine, nitroglycerin, sodium nitroprusside, and nicardipine found no difference between clevidipine, nitroglycerin, and nicardipine in the primary endpoint of death, MI, stroke, or reduced eGFR at 30 days, but clevidipine was more effective in maintaining BP within the prespecified range and was associated with lower mortality than nitroprusside.⁴³ A factor that has limited the widespread use of clevidipine is its relatively high cost.

Fenoldopam mesylate is a selective agonist of dopaminergic 1 receptors located mainly in the renal and splanchnic arteries, with lesser density in the coronary and cerebral arteries.^{4,39,40} IV fenoldopam does not cross the blood-brain barrier and has no central nervous system activity because it is a poorly lipid-soluble molecule and, unlike dopamine, it lowers rather than raises BP. In clinical studies, compared with sodium nitroprusside, fenoldopam demonstrated similar BP-lowering efficacy and beneficial renal effects (increased diuresis, natriuresis, creatinine clearance).⁴⁴ Thus, fenoldopam is mostly useful for BP reduction in patients with renal impairment, those with HF, and those undergoing vascular surgery. Fenoldopam must be administered with caution, if at all, to patients with glaucoma because it increases intraocular pressures.^{4,11,39,40}

For several years, *sodium nitroprusside* was considered the first-choice drug for almost all hypertensive emergencies. It is easily titrated, inexpensive, and has a long record of effectiveness.^{4,11} However, sodium nitroprusside has several drawbacks, including accumulation of toxic metabolites (thiocyanate and cyanide) when used for more than 48 hours, especially in patients with renal or hepatic dysfunction, and the need for invasive BP monitoring and an administration system that protects it from light. High-dose nitroprusside also increases intracranial pressure, obliterates cerebral autoregulation, and reduces regional coronary blood flow. These attributes limit its usefulness in patients with neurologic complications or acute coronary syndromes.^{4,11}

Labetalol is a nonselective α_1 - and β -blocker (in 1:7 ratio) that can be used in many hypertensive emergencies because when given intravenously it has rapid onset of action, potent and sustained effect, and low toxicity. Labetalol reduces peripheral vascular resistance without a reflex increase in systolic volume, and cerebral, renal, and coronary blood flow are maintained. Its main indications are for hypertension associated with aortic dissection, acute coronary syndromes, hypertensive encephalopathy, and adrenergic crisis.^{4,39,40} Labetalol can be used in pregnancy-induced hypertensive crisis because little placental transfer occurs as a result of its negligible lipid solubility.⁴ Labetalol may be used in chronic obstructive pulmonary disease if the patient has no history of an asthmatic component.⁴⁵ *Esmolol* is a β_1 -blocker with an immediate onset and a short duration of action, whose metabolism is not dependent on kidney or hepatic function. Esmolol is used particularly in patients with severe postoperative hypertension, and it can be useful in those with increased cardiac output and heart rate.^{4,39,40}

Table 38.1 provides pharmacologic characteristics and adverse effects of the aforementioned agents, as well as other agents that have been used in the treatment of hypertensive emergencies. Table 38.2 includes a general guide for use of these drugs according to the type of hypertensive emergency.^{4,8,12,39,40}

Treatment of Hypertensive Urgencies

Although hypertensive urgencies are particularly common, high-quality studies on the value of extensive diagnostic testing for target-organ

damage, the need for hospitalization, the type of treatment, and the optimal follow-up in asymptomatic patients with BP elevation are missing.^{46,47} A retrospective cohort study with propensity matching that included all patients presenting with hypertensive urgency (BP > 180/110 mm Hg) to an office in a large health care system in the United States showed that patients sent home compared with those sent to the ED had an equally low risk of major cardiovascular events (0.9%) and equal chance to have uncontrolled hypertension (65% vs. 67%) at 6 months.⁴⁸

Because there is no proven benefit from rapid BP reduction in asymptomatic patients without evidence of acute target-organ damage, current guidelines agree that BP lowering should occur over a longer time than for a hypertensive emergency, and no hospitalization is necessary.^{1,2} BP reduction to less than 160/100 mm Hg may be accomplished within 2 to 4 hours in the ED with resting and oral agents. This may be particularly important for patients without ongoing target-organ damage, who are judged to be at high risk for CV events over the next days because of severe hypertension (e.g., those with known history of aortic aneurysm or repeated episodes of pulmonary edema). However, there are many different types of patients who may present as a hypertensive urgency, the most common of which is the hypertensive patient with low treatment compliance and chronically elevated BP levels. Thus, less aggressive lowering of BP (over several hours to days) is also proposed by some, using strategies such as resumption of antihypertensive therapy (in nonadherent patients), initiation of antihypertensive therapy with long-acting agents (if patients are treatment naive), and treatment modification or addition of another antihypertensive drug (in patients who are already treated). In short, the most important aspect of treatment of hypertensive urgency is not achieving a BP goal but rather ensuring adequate follow-up, generally within 1 week, to an appropriate site of care for chronic hypertension to optimize care and improve chronic BP control.^{1,2,4,11}

Patients with hypertensive urgency should be provided a quiet room in which to rest. An uncontrolled study showed this maneuver resulted in a BP fall of at least 20/10 mm Hg in one-third of these individuals,⁴⁹ and a recent RCT showed that resting produced identical BP reductions to 40 mg of telmisartan over a 2-hour period.²⁴ A large observational study showed that there were no differences in 1-week ED revisit rate and BP control between patients with hypertensive urgencies treated or not with oral medications.⁵⁰ The importance of relaxation is further supported by data suggesting that techniques such as pursed-lip breathing combined with number counting or anxiolytic treatment are associated with greater BP reductions than usual care alone.^{51,52} Another major factor to consider before prescribing antihypertensive medication is assessment of pain. Patients with severe pain not of cardiac or cerebral origin should be given analgesics first to improve it because if they are given acute-acting medications, they could become hypotensive once pain is alleviated.

The choice of drugs for treatment of hypertensive urgencies is much broader than for emergencies because almost all antihypertensives lower BP effectively given sufficient time. Captopril, clonidine, labetalol, and other short-acting drugs have been used most often (Table 38.3).^{4,40,53} A careful history to assess chronic antihypertensive treatment and patient adherence to medication is critical for drug and dose selection, and clinical surveillance is always advisable during the first few hours after drug administration. If a pheochromocytoma is suspected, it is advisable to avoid β -blockers because they can increase BP because of unopposed α -adrenergic activity; this included labetalol because its α -blocking effect is very small. An α -blocker such as doxazosin or prazosin is preferred. Overall, if the patient admits to missing one or more doses of their usual antihypertensive agents, reinstituting these medications in an appropriate order is a reasonable option.

Oral captopril is typically given in a 12.5- to 25-mg dose; angiotensin-converting enzyme (ACE) inhibitors must be used with caution because they can cause or exacerbate renal impairment in the occasional patient

TABLE 38.3 Pharmacologic Agents for Treatment of Hypertensive Urgencies

Drug	Mechanism of Action	Dose	Onset of Action	Duration of Action	Adverse Effects
Captopril	ACE inhibitor	12.5–25 mg PO every 1–2 h	15–30 min	4–6 h	Angioedema, cough, acute renal failure
Clonidine	Central α_2 -agonist	0.1–0.2 mg PO every 1–2 h	30–60 min	6–8 h	Sedation, dry mouth, bradycardia, rebound hypertension after withdrawal
Labetalol	α_1 -, β -Blocker	200–400 mg PO every 2–3 h	30–120 min	6–8 h	Bronchoconstriction, heart block, congestive heart failure
Furosemide	Loop diuretic	20–40 mg PO every 2–3 h	30–60 min	8–12 h	Volume depletion, hyponatremia, hypokalemia
Isradipine	Calcium channel blocker	5–10 mg PO every 4–6 h	30–90 min	8–16 h	Headache, tachycardia, flushing, peripheral edema

PO, By mouth.

Shown are short-acting agents commonly used in the emergency room setting. However, as noted in the text, sometimes longer-acting drugs can be used.

with critical renal artery stenosis.⁵³ Sublingual captopril has a more pronounced BP lowering effect than oral captopril at 10 and 30 minutes but similar at 60 minutes after dosing.⁵⁴ Oral clonidine, 0.1 to 0.2 mg, is one of the most common agents used in this setting. However, patients should not be discharged on clonidine if they have a history of nonadherence to drug regimens because of the risk for rebound hypertension if clonidine is abruptly stopped. Furosemide also can effectively lower BP if elevated pressure is related to volume overload, especially if renal dysfunction is present. However, a common physiologic response of the kidney to elevated BP is natriuresis, so many patients, especially those with normal renal function, are volume depleted, as previously discussed.^{4,53} Further, furosemide is not considered appropriate for treating primary hypertension because of its short duration of action and a rebound in sodium reabsorption if dosed once daily.

Sublingual short-acting nifedipine, although once frequently used, is now contraindicated secondary to a higher incidence of stroke, MI, and death related to precipitous hypotensive episodes after ED release.³² Intermediate-acting dihydropyridine CCBs, such as isradipine and nifedipine, may be used, because their onset of action is between 30 and 90 minutes. Longer-acting CCBs, such as once-daily

nifedipine, sustained-release isradipine, and amlodipine do not have a role in reducing BP in the ED. However, these and long-acting agents from other major antihypertensive classes are valuable tools for long-term BP control, which, as mentioned previously, is the most important aspect of management in these patients.

Despite existing recommendations,^{1,2} a substantial proportion of patients with hypertensive urgency are still treated with IV medications in everyday clinical practice, an intervention that produces similar BP reduction, readmission rates, and adverse outcomes with oral dosing.^{55,56} There are also data suggesting that most of these patients do not receive proper instructions in the ED as proposed in the literature, and providers overestimate how often they refer patients for follow-up, resulting in questionable improvement in long-term outpatient BP control.⁴⁶ Finally, there is a justified concern on the optimal treatment of patients with severe BP elevation and unclear symptoms while they are undergoing workup in the ED to determine whether they have target-organ damage, especially as health system practices and available ED resources vary considerably across the globe.⁵⁷ Further research is needed to clarify these issues and inform decisions for short- and long-term management of patients with hypertensive urgencies.

SELF-ASSESSMENT QUESTIONS

- A patient being evaluated for a routine history and physical is noted to have a BP of 220/130 mm Hg. He states he feels fine except for nightly reflux. Examination shows fundi with atrioventricular nipping and normal heart sounds. Chest radiograph is normal; serum creatinine is 1.3 mg/dL, and urinalysis shows rare red blood cells (RBCs). Management should include which of the following?
 - Immediate hospitalization to the intensive care unit with administration of intravenous labetalol
 - Stat electrocardiogram, cardiac enzymes, and admission to the cardiac care unit
 - Start oral isradipine 5 mg, let the patient rest in a quiet room, and observe for some hours
 - Perform a full workup for secondary causes with plasma metanephrines and catecholamines and aldosterone-renin ratio. Start clonidine, 0.4 mg twice daily.
- A patient presents with an acute hemorrhagic stroke and has a BP of 180/120 mm Hg. You are asked to consult on the patient. Which of the following is the *best* strategy?
 - Give hydralazine (10 mg IV) to bring the BP down as rapidly as possible.
 - Initiate nitroprusside therapy and titrate BP down to 140/90 mm Hg over a 1-hour period.
 - Start nicardipine infusion and reduce BP to 160/90 mm Hg within an hour.
 - Do not treat the hypertension because it may worsen the stroke.
- A patient presents with acute chest pain. On examination, BP in the right arm is 190/120 mm Hg but in the left arm is only 160/90 mm Hg. The chest radiograph shows a widened mediastinum. What is the *best* initial treatment to lower the BP?
 - Start a nitroprusside infusion, targeting BP of 140/90 mm Hg in the right arm.
 - Start a labetalol infusion, targeting BP of 120 mm Hg systolic in the right arm.
 - Start an esmolol infusion, targeting BP of 120 mm Hg systolic in the left arm.
 - Do not treat the BP because it is important to preserve cerebral perfusion.

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